ENGLISH PAPER SUMMARY | ALL ORIGINAL NUMBERED FIGURES

Simulating ice-wave interactions in the Laurentian Great Lakes using a fully coupled hydrodynamic-ice-wave model

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Two-way wave-ice coupling produces localized breakup and ice-edge changes, but improvements over an earlier Great Lakes model largely also depend on initialization, forcing, and calibration. These effects must not be conflated.

Prepared independently from the original paper. This is a new English synthesis, not a translation of the Korean notes. The figure atlas reproduces all 18 numbered figures with explanatory reading notes. Section and equation references refer to the source publication.

Question and comparison design

The study extends FVCOM-SWAVE-UG-CICE with wave-induced ice breakage and floe-size-dependent lateral melt, in addition to ice-induced wave attenuation. It evaluates whether direct two-way interaction changes Great Lakes ice and waves during winter 2011, and compares uncoupled, one-way, and fully coupled configurations. [Sections 1-2; Figures 1-2]

Runs R1, R2, and R3 represent no direct ice-wave coupling, ice-to-wave attenuation only, and attenuation plus wave-to-ice breakup. R4 regenerates an earlier Bai et al. configuration. The R2-R3 difference isolates the added direct breakup mechanism more closely than R4-R3, which also includes changes to calibration and initialization. [Table 1]

Mechanism: from spectrum to breakup to melt

The wave action balance scales wind input, whitecapping, and breaking by open-water fraction and adds concentration-weighted ice attenuation. The selected IC4M6 coefficient is a piecewise function of frequency. This attenuation component does not itself require a learned floe size distribution. [Section 2.2; Equations 3-5]

Breakup is triggered when significant wave-induced strain exceeds a critical strain. The wave spectrum, ice thickness, transmission, and ice-wave dispersion enter the strain calculation. Freshwater material assumptions set brine volume to zero. Passing the threshold causes a representative floe diameter to be updated using a parameterized broken-floe distribution. [Sections 2.3.1-2.3.2]

Smaller floes have greater perimeter relative to area, increasing lateral melt through a term proportional to inverse representative diameter. Breakage therefore affects ice concentration through subsequent thermodynamic evolution; splitting a floe does not by itself destroy its ice area or volume. The model uses a representative diameter and prescribed distributional assumptions, rather than WIFF1.0's learned 12-bin fracture histogram. Wave radiation stress acting on floe sides is not included. [Section 2.3.3]

Forcing, validation, and calibration

NARR supplies meteorological forcing; wind speeds are increased by 20% in the proposed configuration. GLSEA surface temperature supports ice initialization, and lateral/basal melt coefficients and ice strength are adjusted. NIC ice concentration and Lake Erie in situ ice observations provide validation. Satellite products have cloud, resolution, and processing uncertainty. [Sections 2.4-2.5 and Appendix A]

The study explicitly examines initialization, wind forcing, melt rates, temperature perturbations, and ice strength. Figures 14-18 are part of the paper's evidence, not optional illustrations: they show why a better final simulation cannot be attributed solely to newly added coupling.

What coupling changes

In February, the ice concentration and thickness maps are similar across the main configurations, while including attenuation strongly suppresses unrealistic waves in frozen regions. The no-coupling model produces approximately 0.7 m significant wave height in ice-rich Lake Erie, whereas the ice-aware cases produce negligible waves there. [Section 3.1; Figure 6]

March averages show relatively modest additional differences from direct breakup. Short-lived and localized changes are more pronounced: an illustrated event has ice concentration reductions up to about 10% near marginal ice, and the 80%-concentration edge in Whitefish Bay shifts by as much as approximately 3.5 km in the direction of wave propagation. These are event-specific modeled effects, not a uniform basin-wide retreat. [Section 3.2; Figures 7-11]

Hydrodynamic-wave coupling also permits indirect effects on thermal structure. Thus the absence of a direct wave-to-ice connection does not imply that every ice quantity must be exactly identical across all configurations. The paper distinguishes this weak indirect pathway from the added direct breakup process. [Section 3.2.1]

Regional skill and attribution

For January-May 2011, Table 2 reports fully coupled mean ice cover of 36.9% in Huron, 5.8% in Ontario, 14.1% in Michigan, 60.6% in Erie, and 8.9% in Superior, compared with NIC values of 30.2%, 6.7%, 12.5%, 49.0%, and 11.2%. These are seasonal lake averages and should not replace spatial or event-level diagnostics.

The earlier-model relative ice-cover error in Superior is +137.5%, versus -20.0% for the proposed fully coupled model; Ontario changes from -49.6% to -13.1%. The reduction in absolute relative error is 117.5 and 36.5 percentage points, respectively. These are not percentages of additional accuracy due to breakup. One-way and fully coupled seasonal averages remain close. [Section 3.3 and Table 2]

At Lake Erie station 4a, both updated one-way and fully coupled models better reproduce the February retreat than the earlier configuration. The paper attributes most of this improvement to calibration and ice initialization; direct breakup causes only a small additional change there. This qualifies the earlier hypothesis that missing breakup was the primary explanation for the Bai et al. event error. [Section 3.4; Figure 13]

Limitations and implications for surrogate evaluation

The results concern a particular winter, parameter set, grid, and observation network. Omitted river inflows, cold spring forcing, satellite uncertainty, and neighborhood-scale variability complicate the interpretation of melting and retreat. A representative-floe closure is not a direct validation of the full observed floe size distribution.

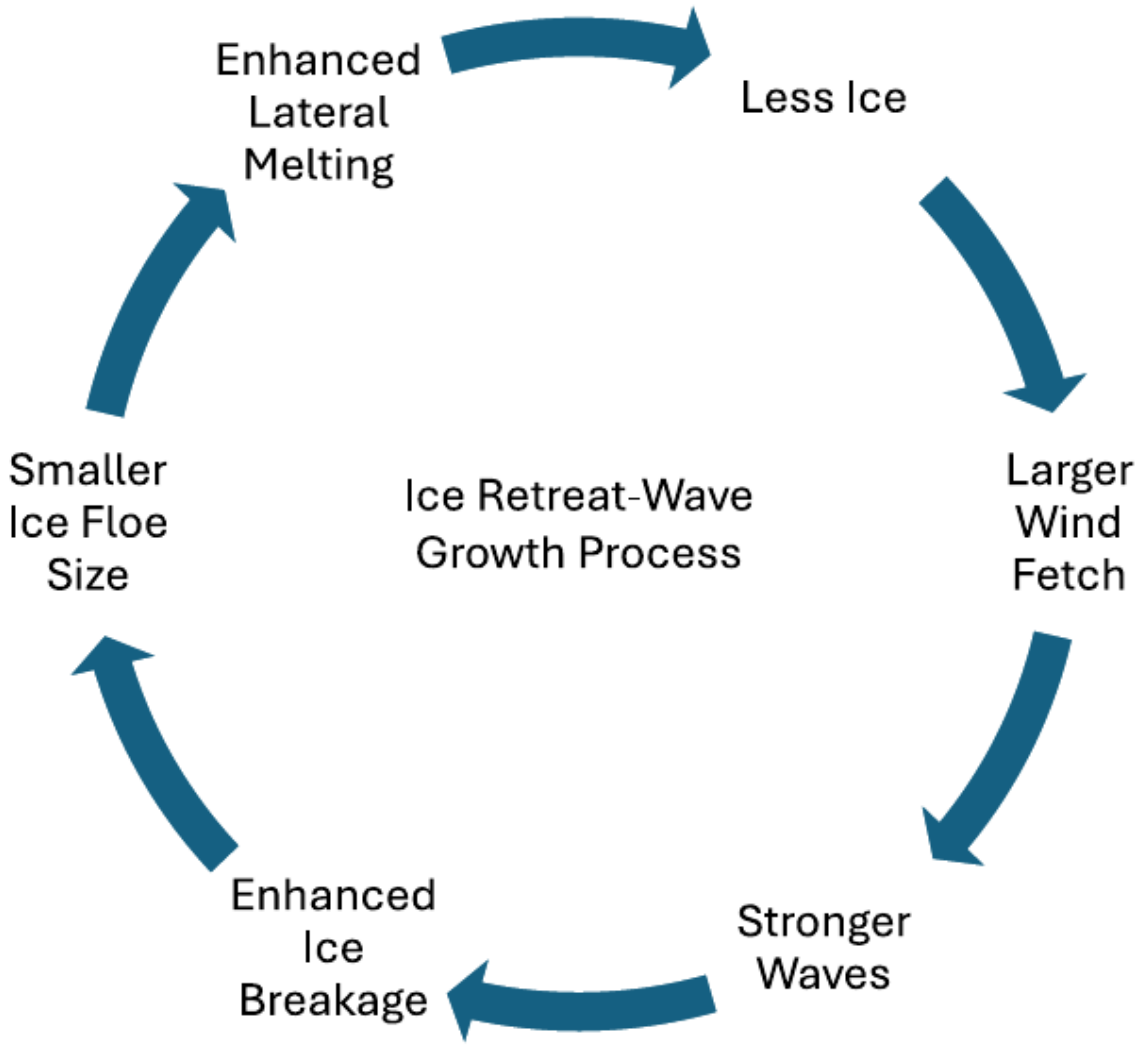
For the Great Lakes project, preserve matched baselines when testing a surrogate: hold forcing, initialization, and calibration fixed when comparing numerical and learned breakup. Evaluate local edge displacement and short events in addition to seasonal ice cover. The paper's ablations supply a useful template for separating changed physics from changed input quality; it does not itself develop an ML surrogate.

Source and figure attribution

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FIGURE ATLAS | 01 / 18

Figure 1. The proposed ice-retreat/wave-growth feedback



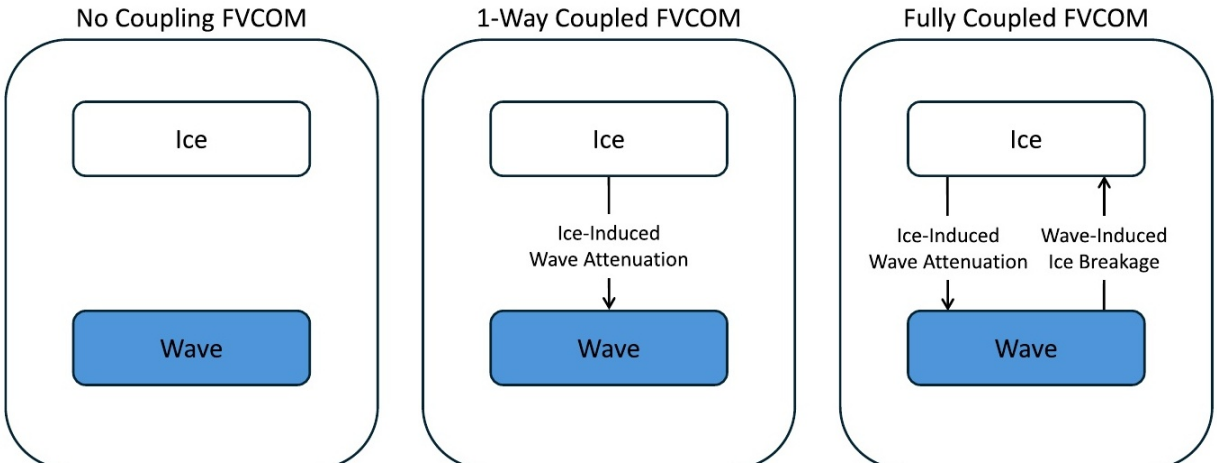
How to read this figure

The conceptual loop links less ice, increased fetch, stronger waves, ice breakup, smaller floes, and enhanced lateral melt. It describes a possible positive feedback rather than a measured rate law. Each link requires its own assumptions; breakup affects melt through floe geometry instead of immediately removing ice mass.

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FIGURE ATLAS | 02 / 18

Figure 2. Three coupling architectures



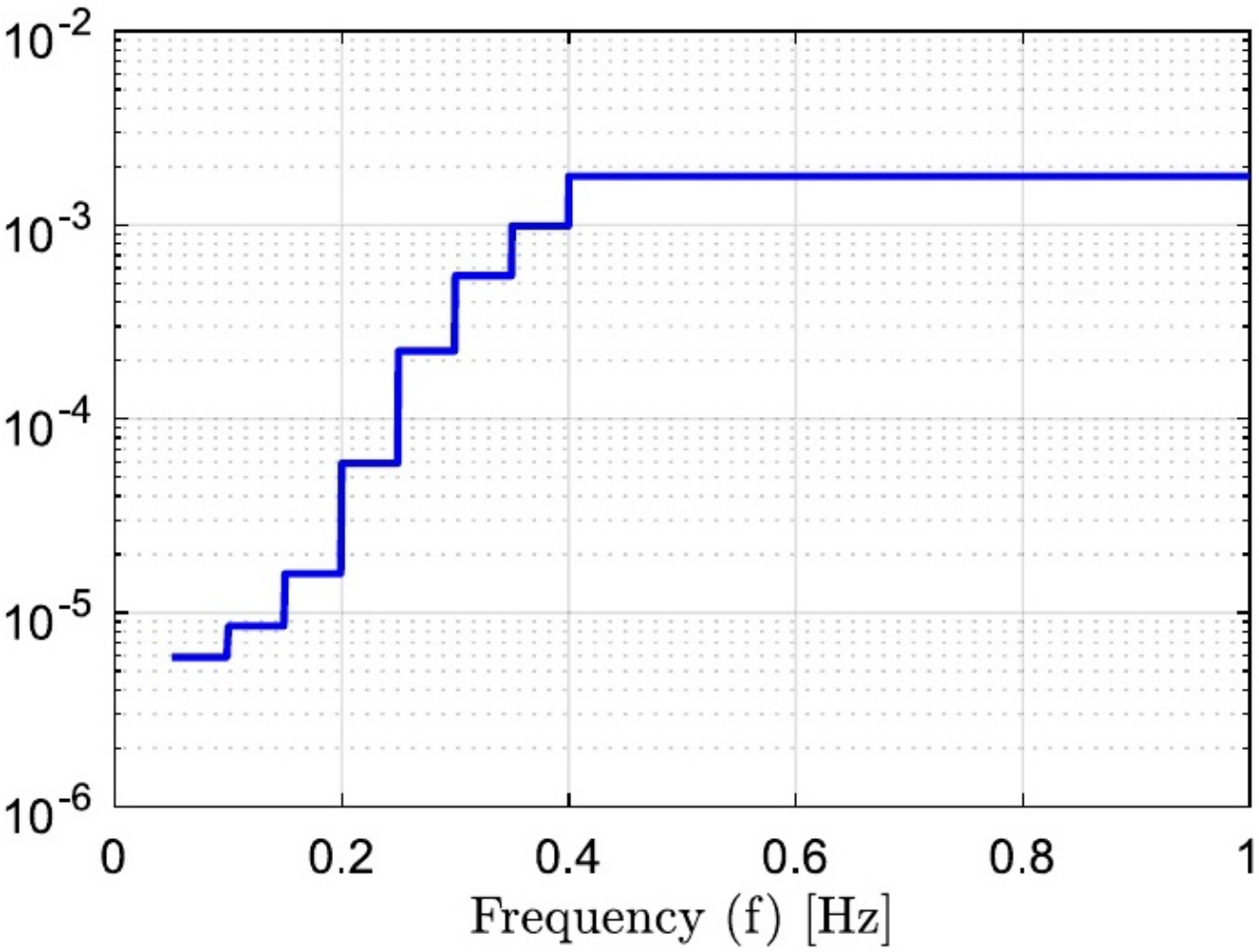
How to read this figure

The diagram distinguishes no direct ice-wave interaction, ice-to-wave attenuation, and two-way attenuation plus breakup. It provides the basis for interpreting matched comparisons. The earlier-model R4 comparison also involves other configuration differences and should not be read as a clean coupling-only experiment.

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FIGURE ATLAS | 03 / 18

Figure 3. IC4M6 attenuation coefficient



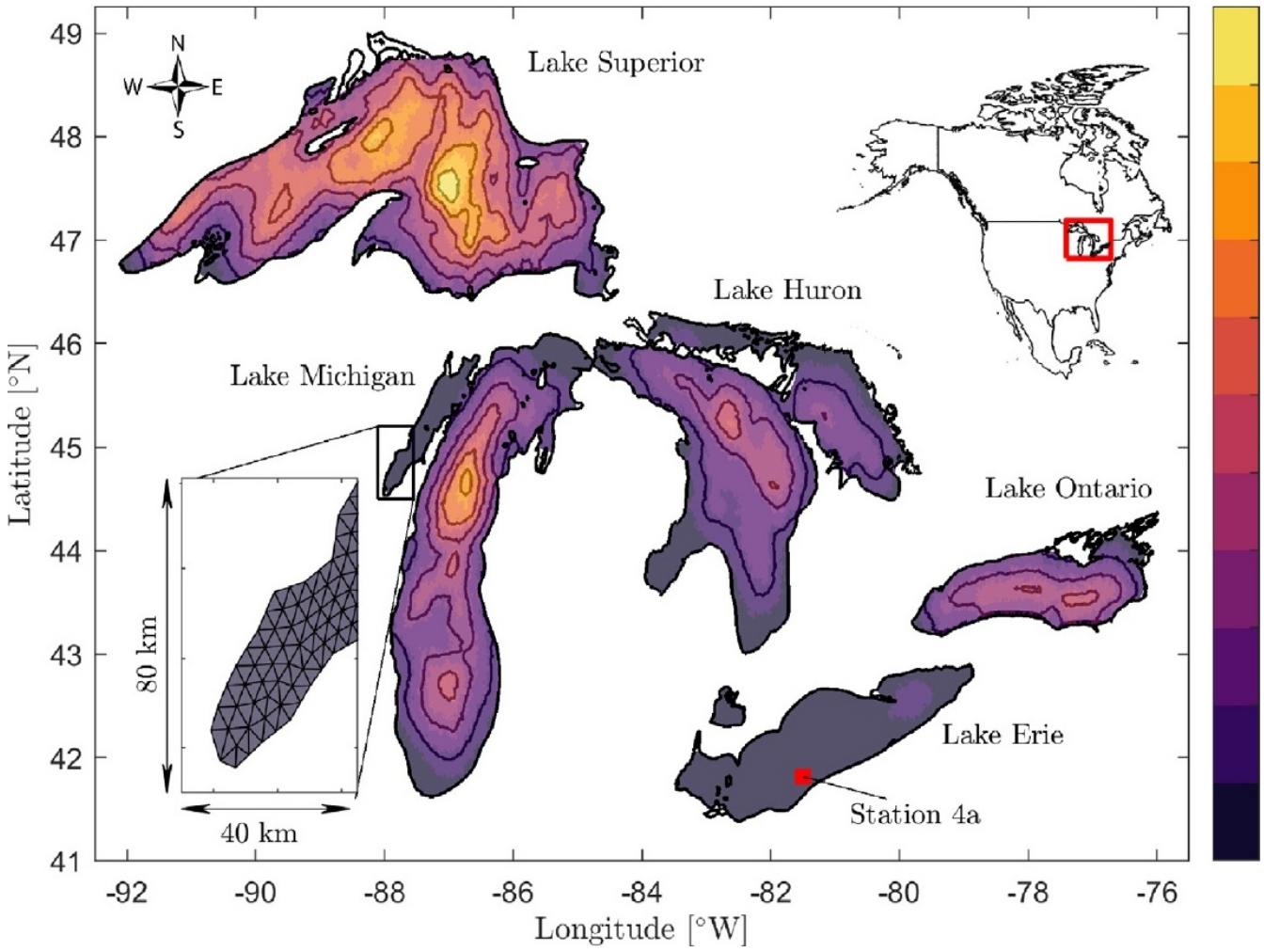
How to read this figure

The stepwise coefficient increases with wave frequency, so short-period components are more strongly damped in the selected formulation. This figure documents the wave sink used in the experiments; it is separate from the strain threshold and floe-diameter update controlling breakup.

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FIGURE ATLAS | 04 / 18

Figure 4. Five-lake bathymetry and unstructured grid



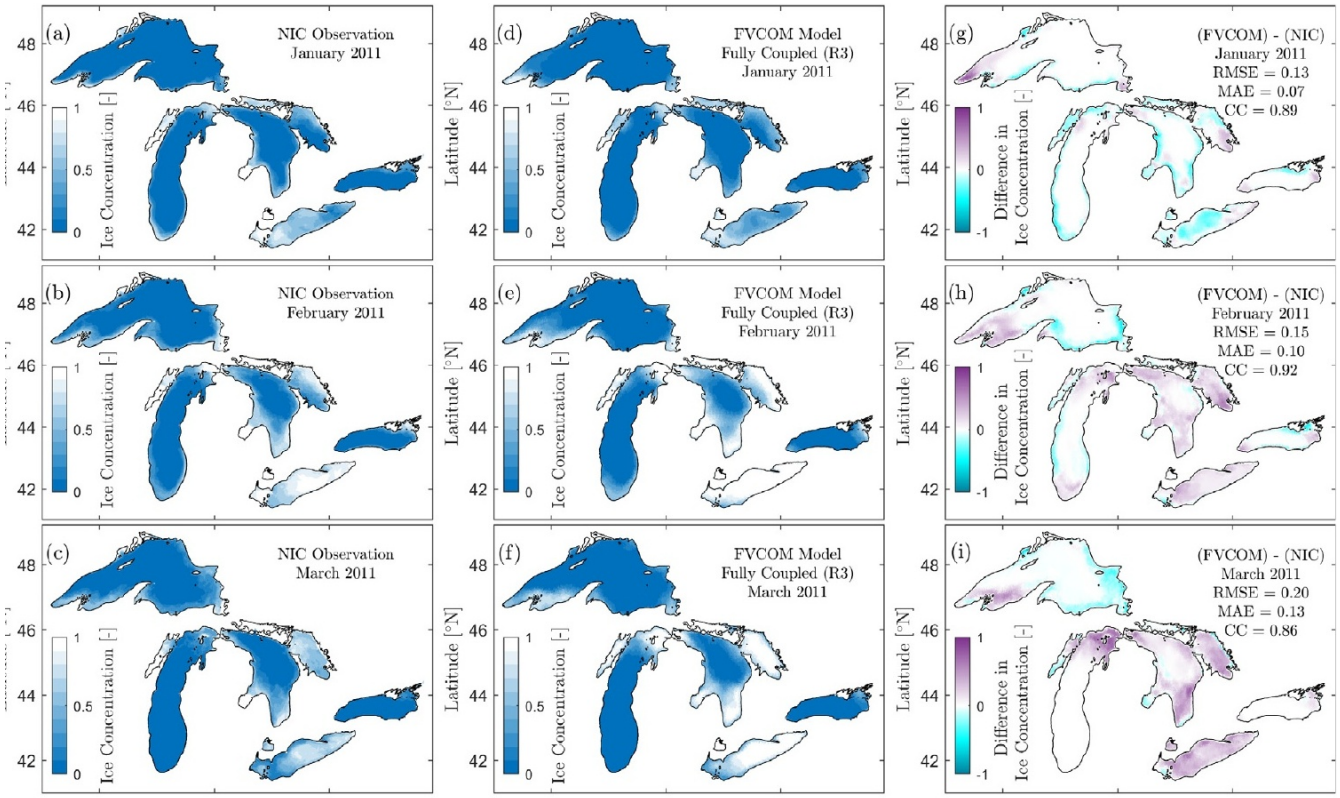
How to read this figure

The map locates the Great Lakes, illustrates mesh structure in Green Bay, and identifies Lake Erie station 4a. Complex geometry motivates the unstructured-grid framework. The point-validation site cannot by itself constrain every spatial feature of lake-wide ice and wave fields.

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FIGURE ATLAS | 05 / 18

Figure 5. Monthly ice concentration against NIC



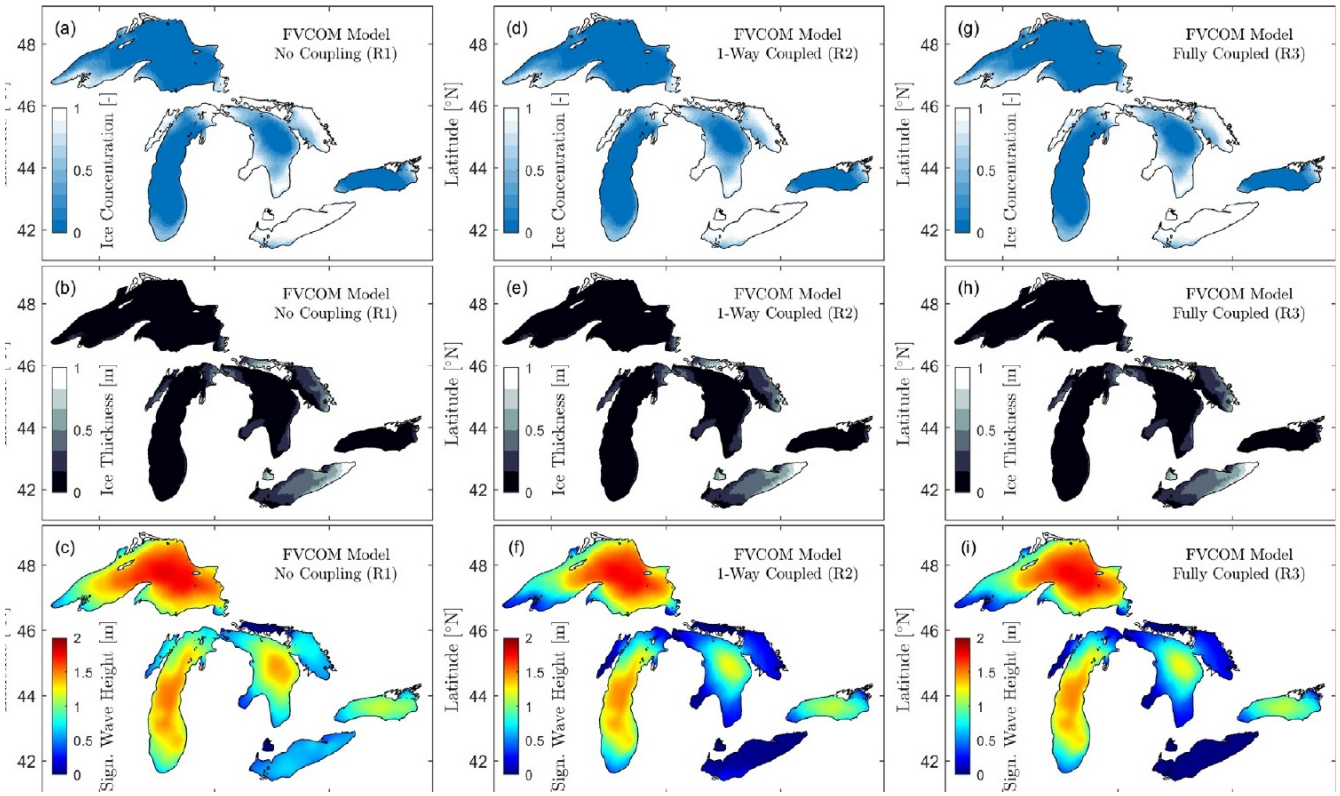
How to read this figure

Observed and modeled ice concentrations for winter 2011 are paired with spatial differences and skill statistics. The model captures broad patterns but retains regional timing and melt discrepancies. These comparisons assess the full configured system, including its forcing and calibration.

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FIGURE ATLAS | 06 / 18

Figure 6. February averages for the coupling experiments



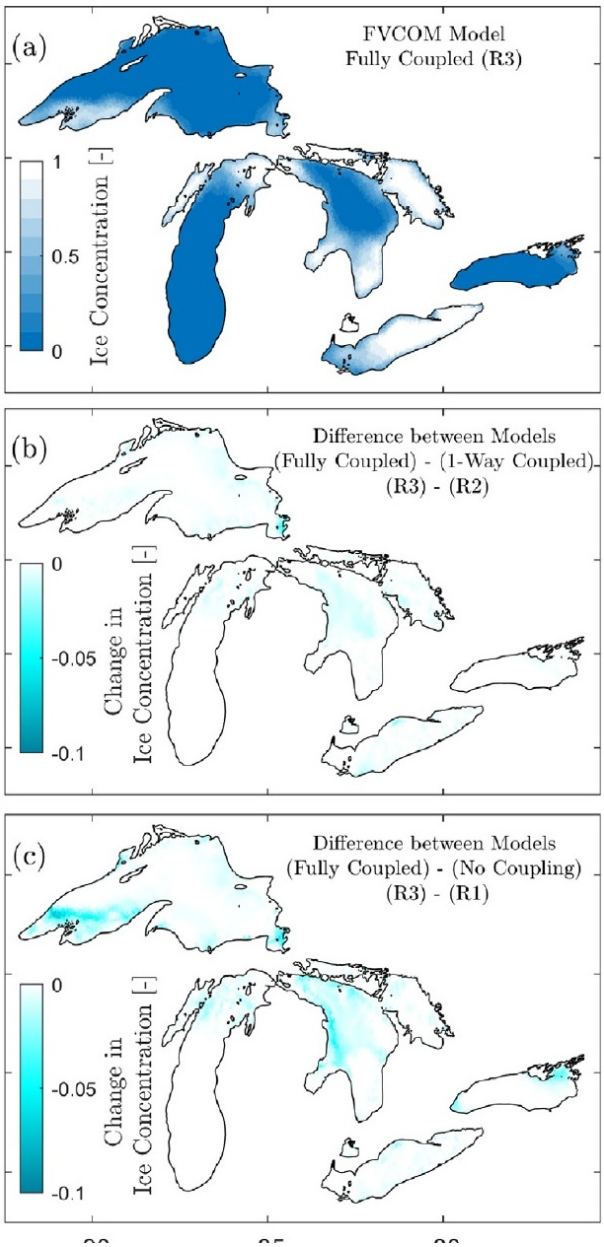
How to read this figure

Ice concentration, thickness, and wave height are compared across the three main coupling levels. Similar ice maps coexist with very different wave fields: attenuation removes unrealistic wave energy under dense ice. The additional breakup effect is relatively small in these monthly averages.

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FIGURE ATLAS | 07 / 18

Figure 7. March ice-concentration differences



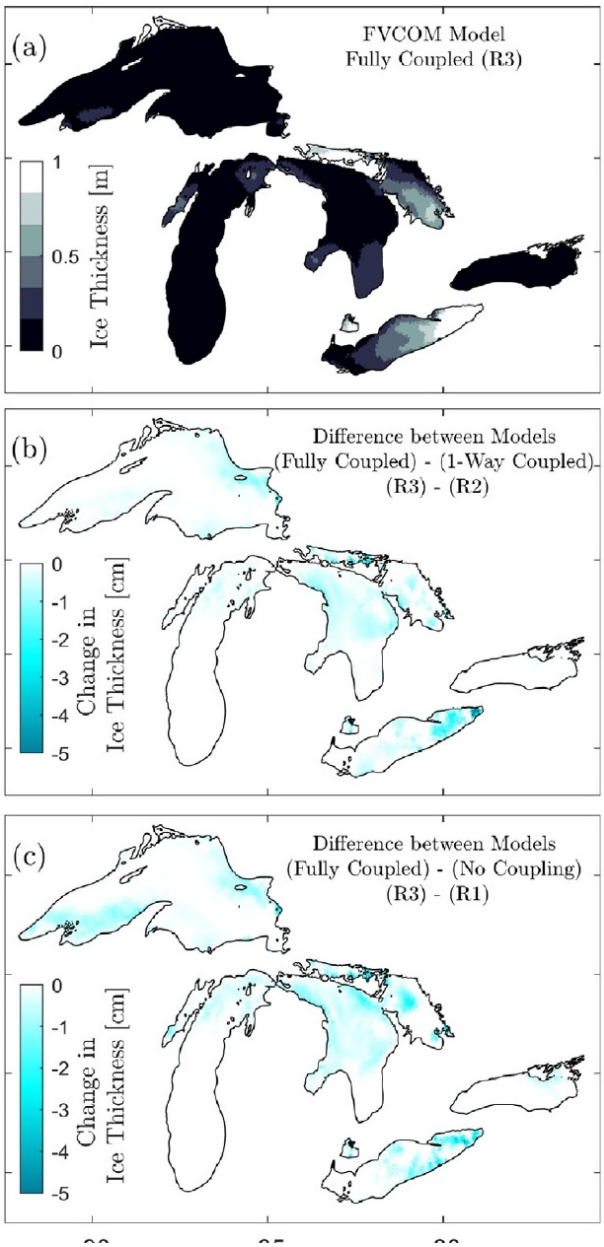
How to read this figure

The fully coupled concentration field and its differences from the one-way and no-coupling experiments highlight marginal-ice changes. Monthly averaging smooths short-lived breakup responses; these maps should be read alongside the instantaneous and local examples rather than as their replacement.

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FIGURE ATLAS | 08 / 18

Figure 8. March ice-thickness differences



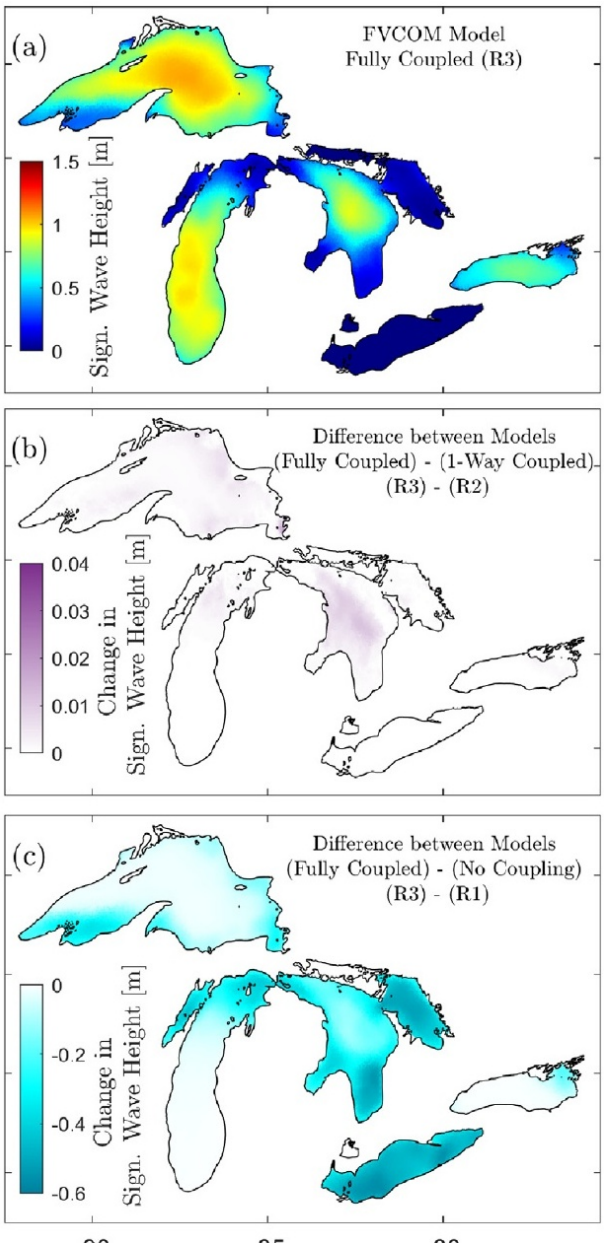
How to read this figure

The corresponding thickness maps show comparatively small changes between configurations. In the modeled feedback, altered floe sizes affect thermodynamic evolution. A lower modeled thickness should not be interpreted as the instantaneous mechanical consequence of splitting an otherwise unchanged floe.

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FIGURE ATLAS | 09 / 18

Figure 9. March wave-height differences



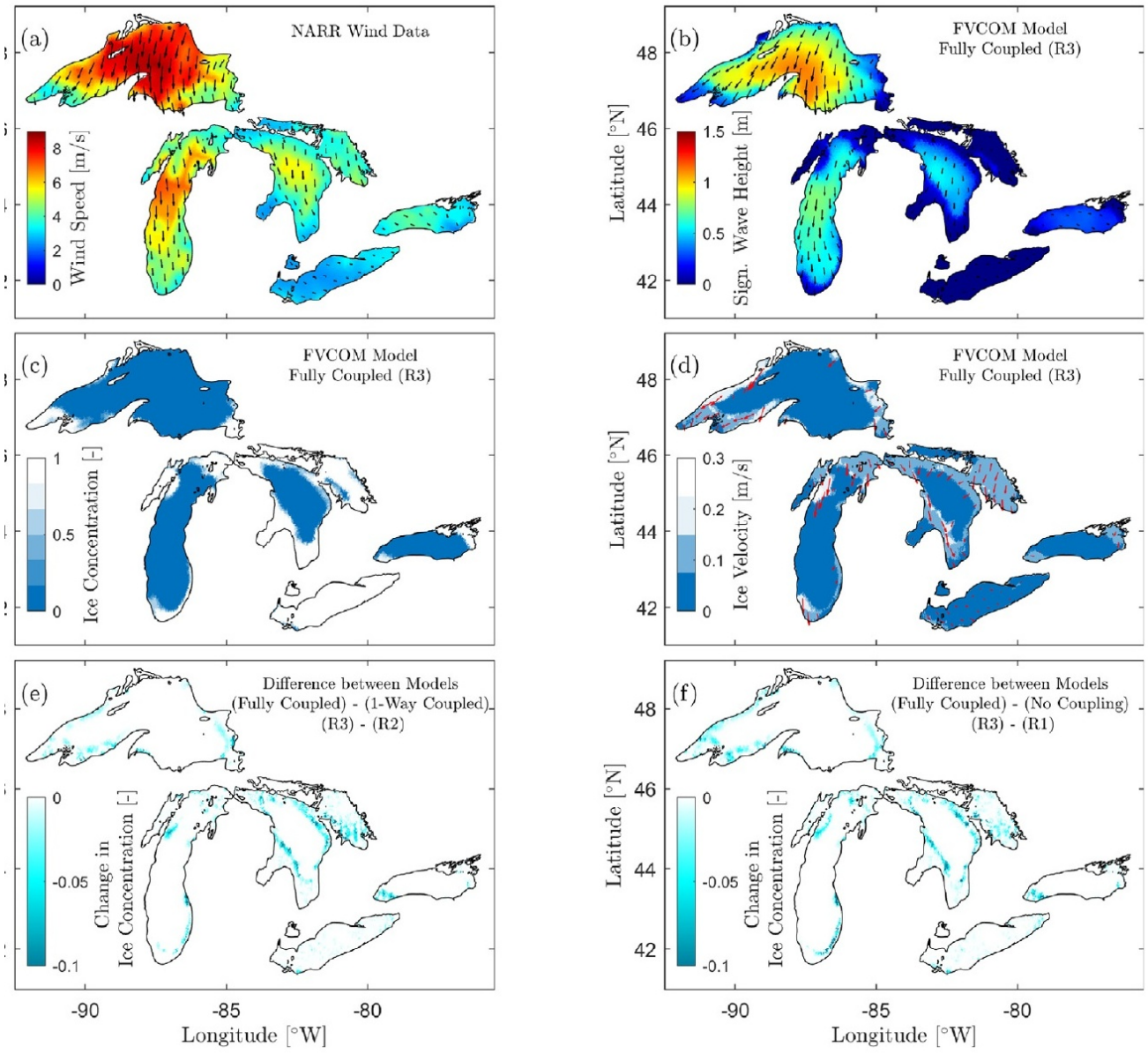
How to read this figure

Wave heights respond strongly to including ice attenuation relative to the no-coupling case. The fully coupled versus one-way difference is much smaller and is associated with changed ice conditions. This contrast separates the dominant attenuation effect from the incremental feedback through breakup and melt.

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FIGURE ATLAS | 10 / 18

Figure 10. A daily event on 30 January 2011



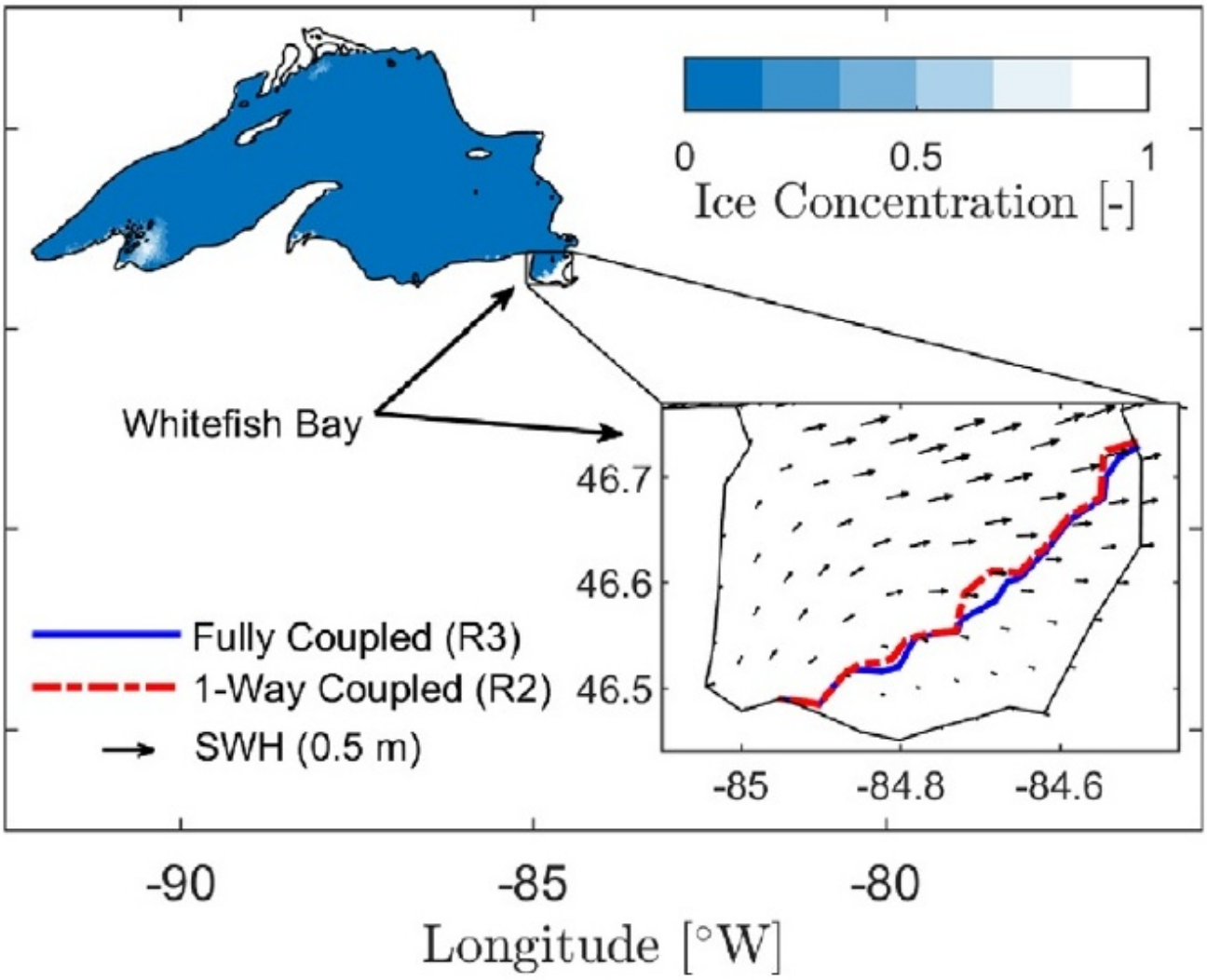
How to read this figure

Wind, waves, ice concentration, ice velocity, and coupling differences show where short-term effects emerge. Reductions near marginal ice can be more visible than in monthly maps. The example supports localized coupling sensitivity, not a basin-wide claim of a uniform 10% reduction.

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FIGURE ATLAS | 11 / 18

Figure 11. Whitefish Bay ice-edge displacement



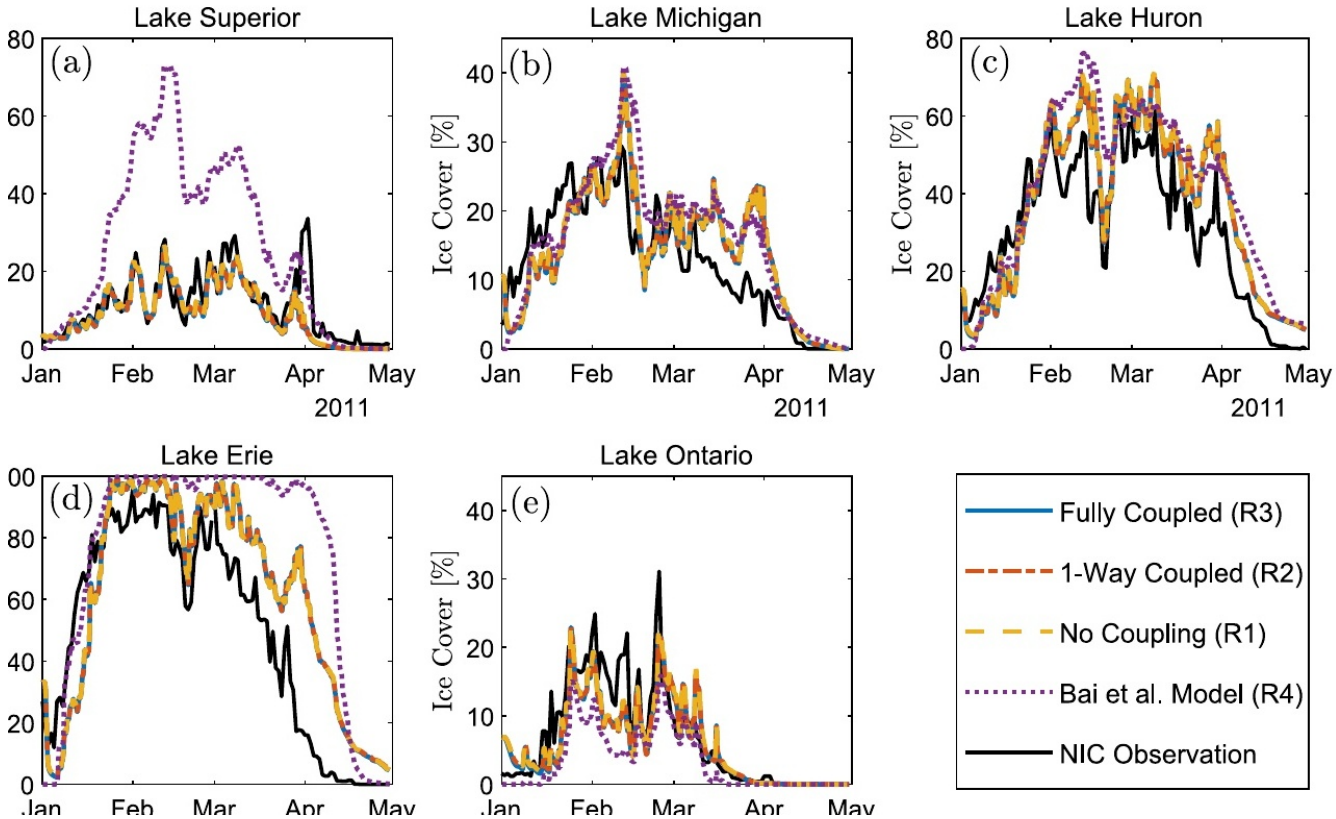
How to read this figure

At noon on 3 April 2011, the 80% ice-concentration contour differs between fully and one-way coupled runs, with the largest separation around 3.5 km. Wave-direction arrows connect the displaced edge with the modeled propagation direction. This is a specific model event, not a general observed retreat distance.

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FIGURE ATLAS | 12 / 18

Figure 12. Daily lake-averaged ice cover



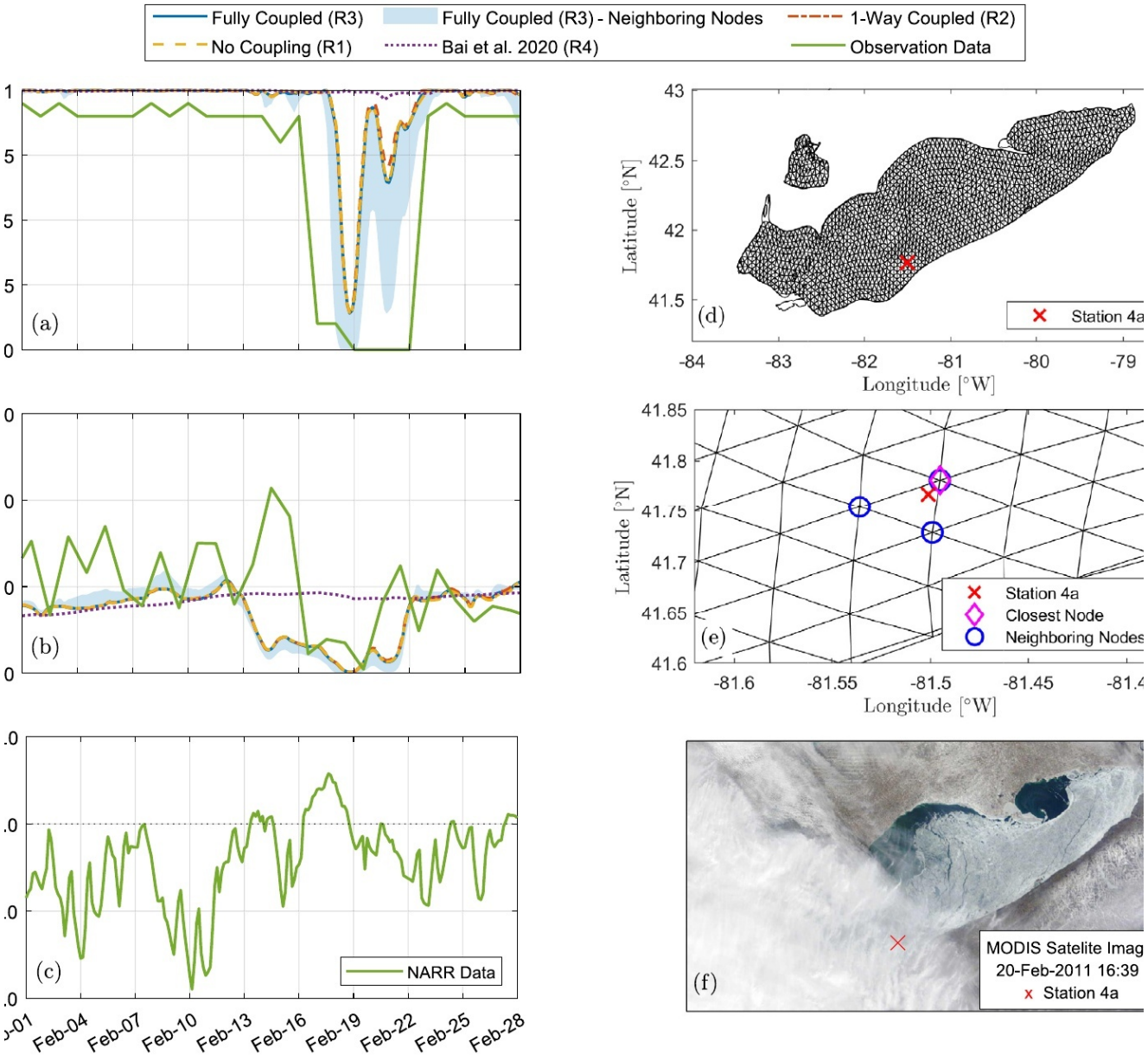
How to read this figure

Time series compare the three main configurations and the earlier-model regeneration with NIC for January-May 2011. Updated one-way and fully coupled curves are close, while both can improve on the earlier setup. Attribution therefore requires the initialization and calibration experiments that follow.

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FIGURE ATLAS | 13 / 18

Figure 13. Station 4a: ice, temperature, and spatial context



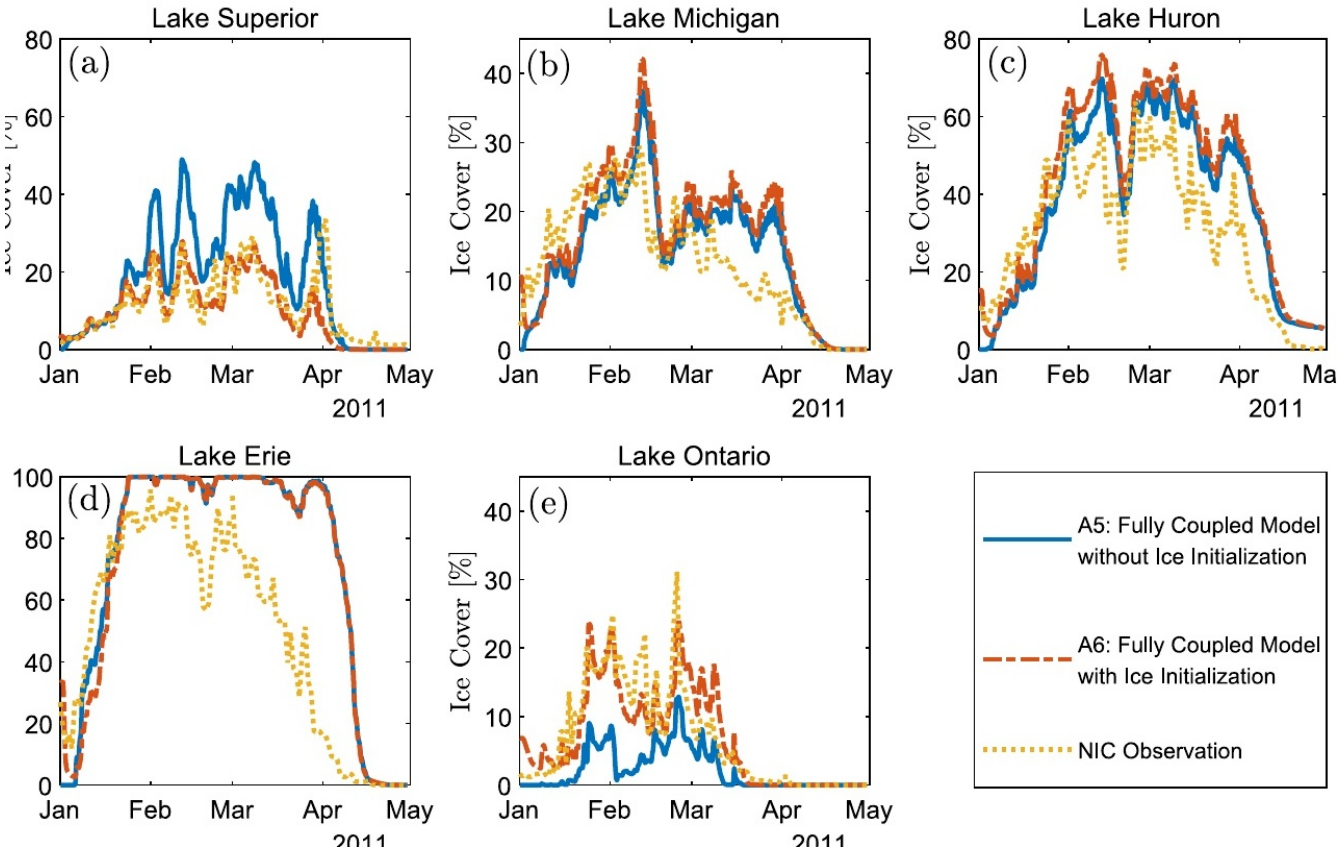
How to read this figure

Ice concentration and thickness time series are shown with atmospheric temperature, neighboring grid nodes, and a satellite scene. Improvements over the earlier configuration occur in both updated coupling levels. The example also illustrates cloud and spatial-support uncertainty in observational comparisons.

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FIGURE ATLAS | 14 / 18

Figure 14. Sensitivity to ice initialization



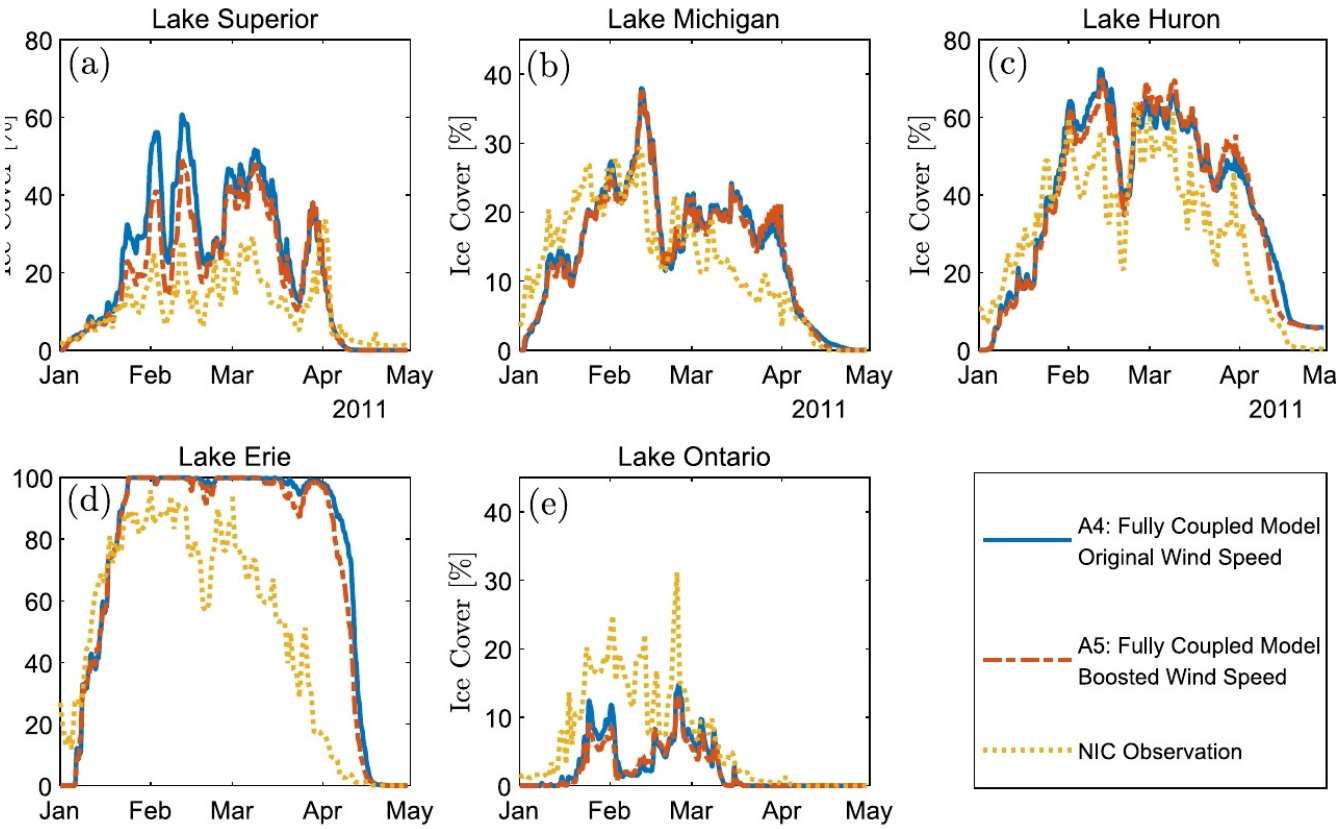
How to read this figure

With other settings matched as described in the paper, initializing ice changes the seasonal ice-cover trajectory, especially in Superior and Ontario. The persistent influence explains why initialization must be held fixed when estimating the value of new breakup physics or a surrogate.

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FIGURE ATLAS | 15 / 18

Figure 15. Sensitivity to stronger wind forcing



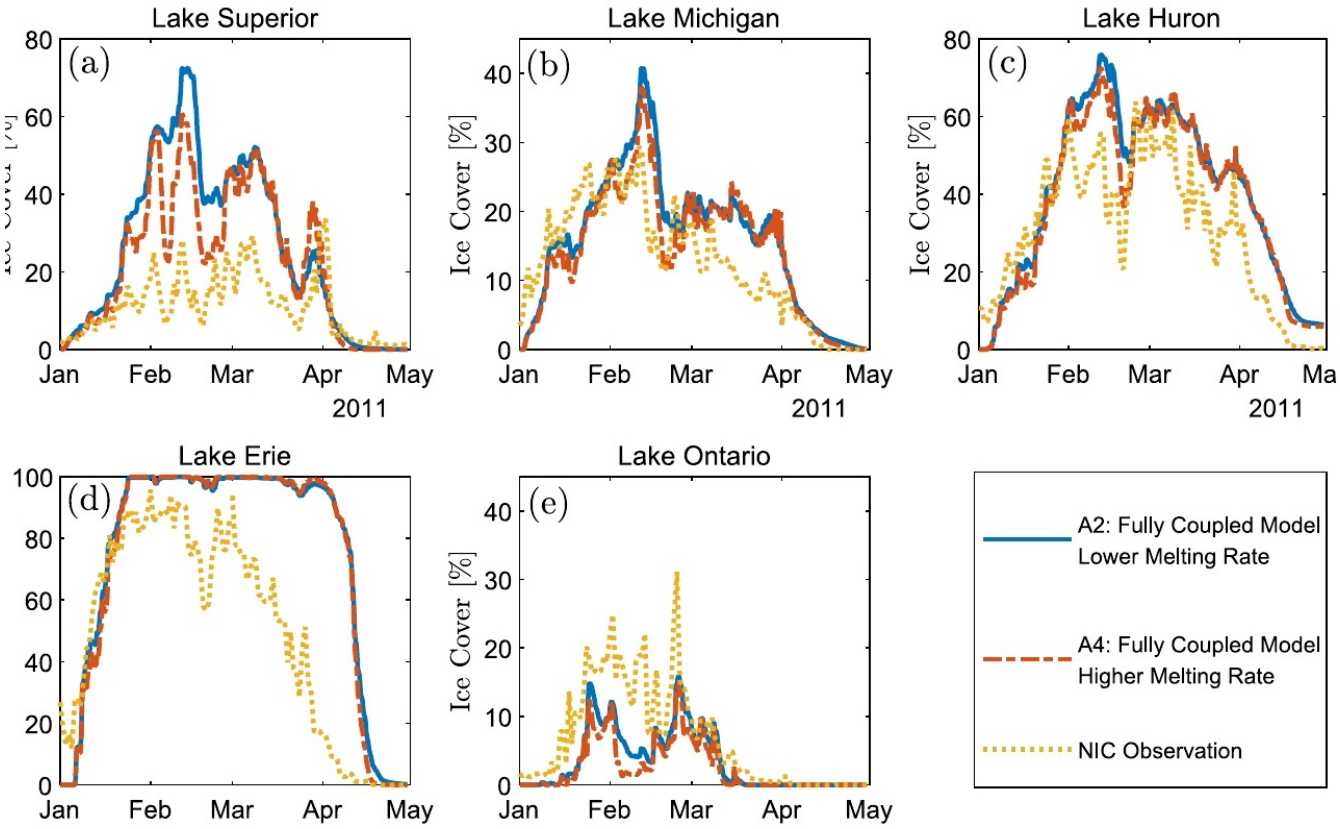
How to read this figure

Boosting NARR winds alters modeled lake ice evolution, with a notable response in Superior. This is a forcing experiment rather than a coupling experiment. It demonstrates that an apparent improvement in the final system can arise from meteorological corrections.

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FIGURE ATLAS | 16 / 18

Figure 16. Sensitivity to melt coefficients



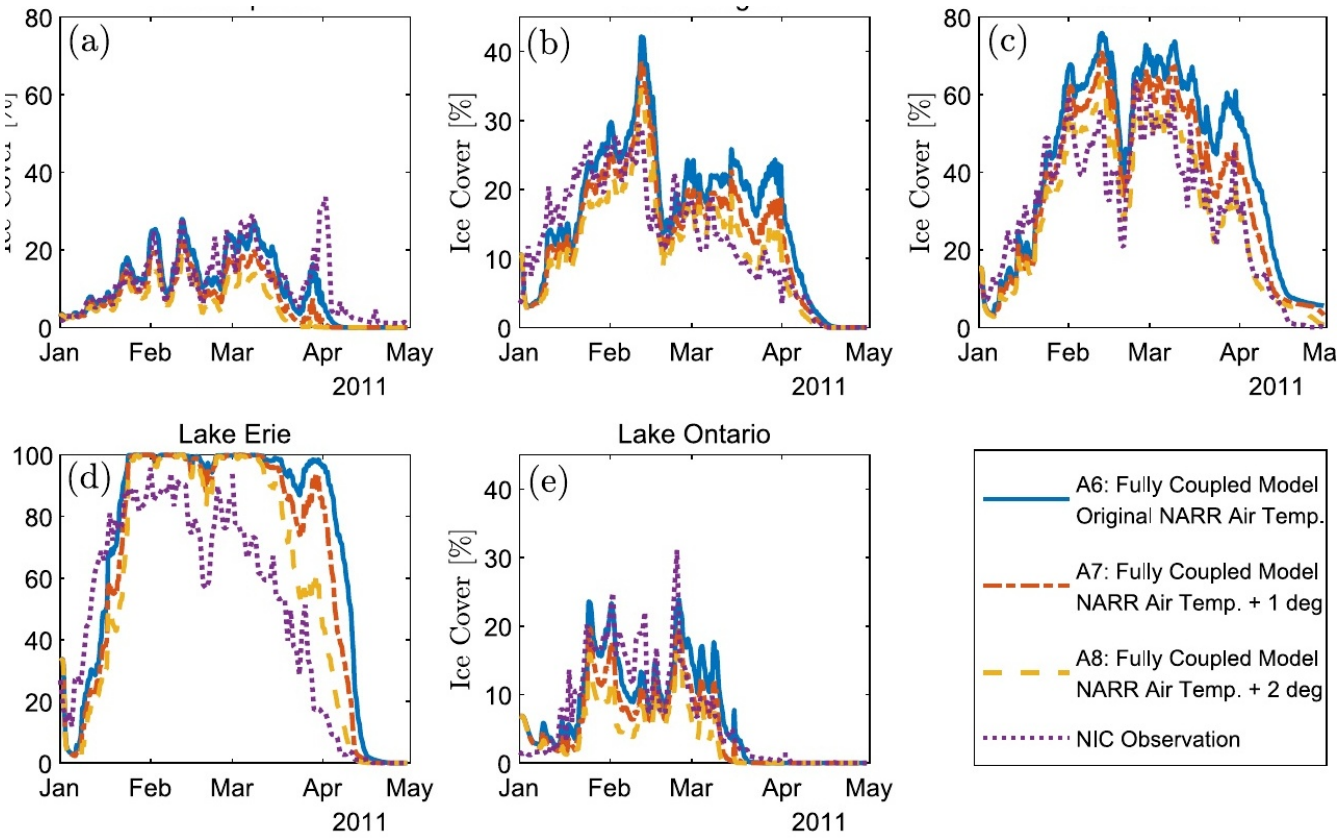
How to read this figure

Higher lateral and basal melting rates alter the seasonal ice-cover curves. The response exposes uncertainty in thermodynamic tuning and its interaction with floe geometry. A surrogate evaluated against only one calibrated numerical reference would inherit those reference-model assumptions.

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FIGURE ATLAS | 17 / 18

Figure 17. Sensitivity to air temperature



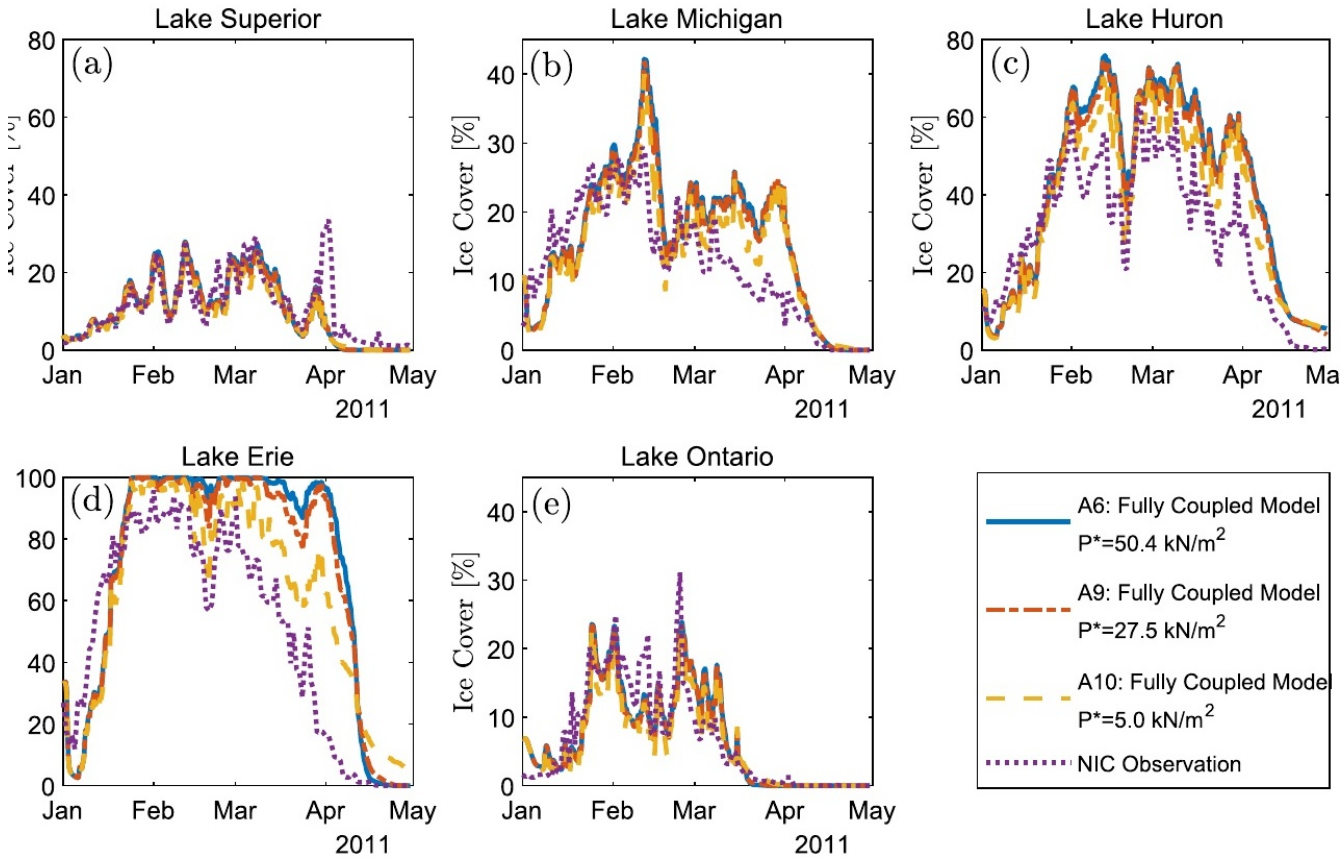
How to read this figure

Warmer forcing reduces ice cover in the tested perturbations. The main runs retain original NARR temperature because biases differ among lakes. This experiment cautions against a single basin-wide temperature correction and against attributing all melt discrepancies to wave-driven fracture.

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FIGURE ATLAS | 18 / 18

Figure 18. Sensitivity to ice strength



How to read this figure

Different ice-strength parameters change the predicted ice-cover histories, with notable responses in Erie, Michigan, and Huron. The sensitivity is another source of uncertainty beyond the wave sink and breakup closure. It helps motivate controlled, matched comparisons in future Great Lakes experiments.

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